

Intro to App Development

Aidan Melvin



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Plan

- Overview
- Software development landscape
- Mobile app development
 - Technologies
 - SDLC
 - Code Samples
 - Discussion
 - Next Steps

Software Dev Landscape

- Extremely broad!
 - Web & mobile, distributed systems, ML, embedded, DevOps, etc.
- Software models
 - SaaS (Software as a Service)
 - Examples: Google Workspace, Notion, Figma
 - Characteristics: subscription-based, scalable, UX-focused
 - PaaS (Platform as a Service)
 - Examples: Heroku, Firebase, Vercel
 - Focus: developer productivity, abstracting infra
 - IaaS (Infrastructure as a Service)
 - Examples: AWS, GCP, Azure
 - Focus: flexibility, scalability, infrastructure control

What?



What is app development?

- Writing code that runs on another device!



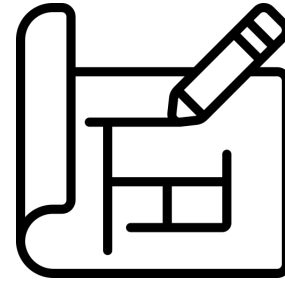
Surprising things?

- Many frameworks, anyone can get started
 - Collaborative effort
- Often need to adapt for all users and devices

Why?

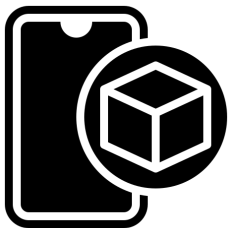


- High demand
- Transferable skills
 - Good pay
 - Flexibility
- Global Market
- Job security



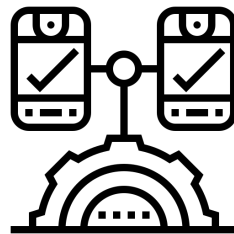
- Creativity
- Personal Growth
 - User Impact
- Entrepreneurial opportunities
 - Networking
 - Fun & Fulfilling

Getting Started

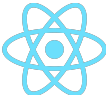


Native

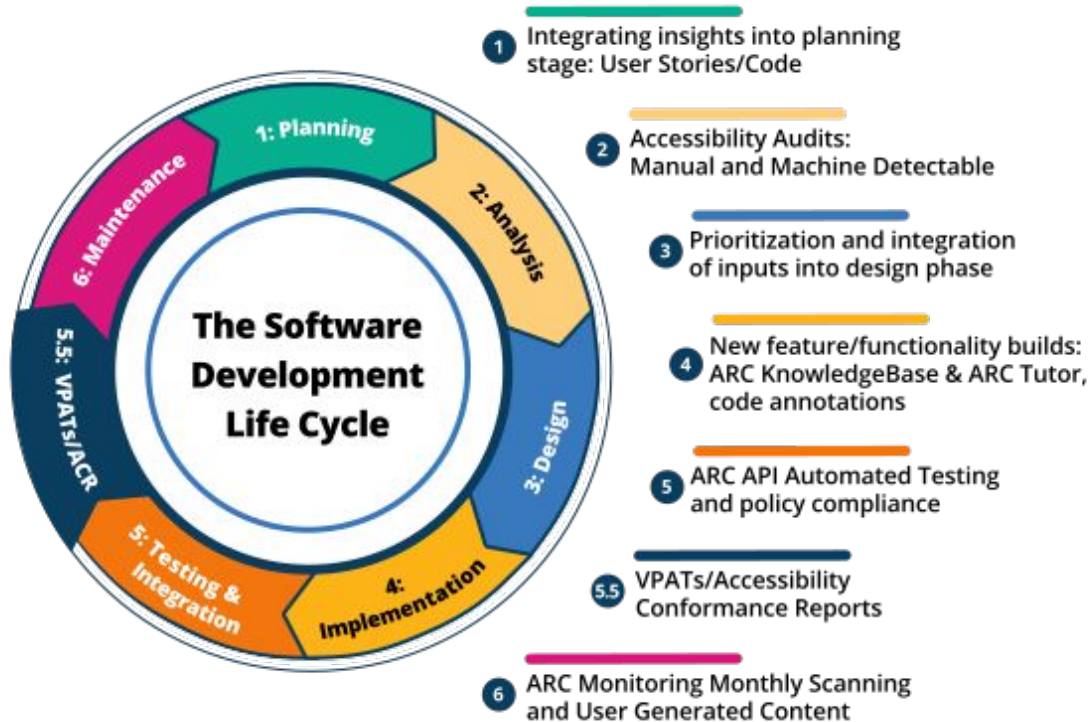
- SwiftUI 
- Android 



Cross Platform

-  • React Native
-  • Flutter
-  • Xamarin

Example: Travel Itinerary



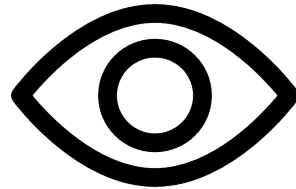
https://github.com/appdevumd/bitcamp_reactnative

https://github.com/appdevumd/bitcamp_flutterdemo

Planning

Step 1: Plan

Essential step in app development. Define purpose, audience, features, and functionality. Map out user journey and identify challenges. Minimizes errors, reduces costs, improves quality.

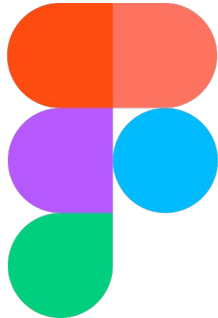


View locations

Designing

Step 2: Design

Essential aspect of app development. Create appealing and user-friendly interface. Select colors, images, and layout.
Enhances usability and improves app satisfaction.

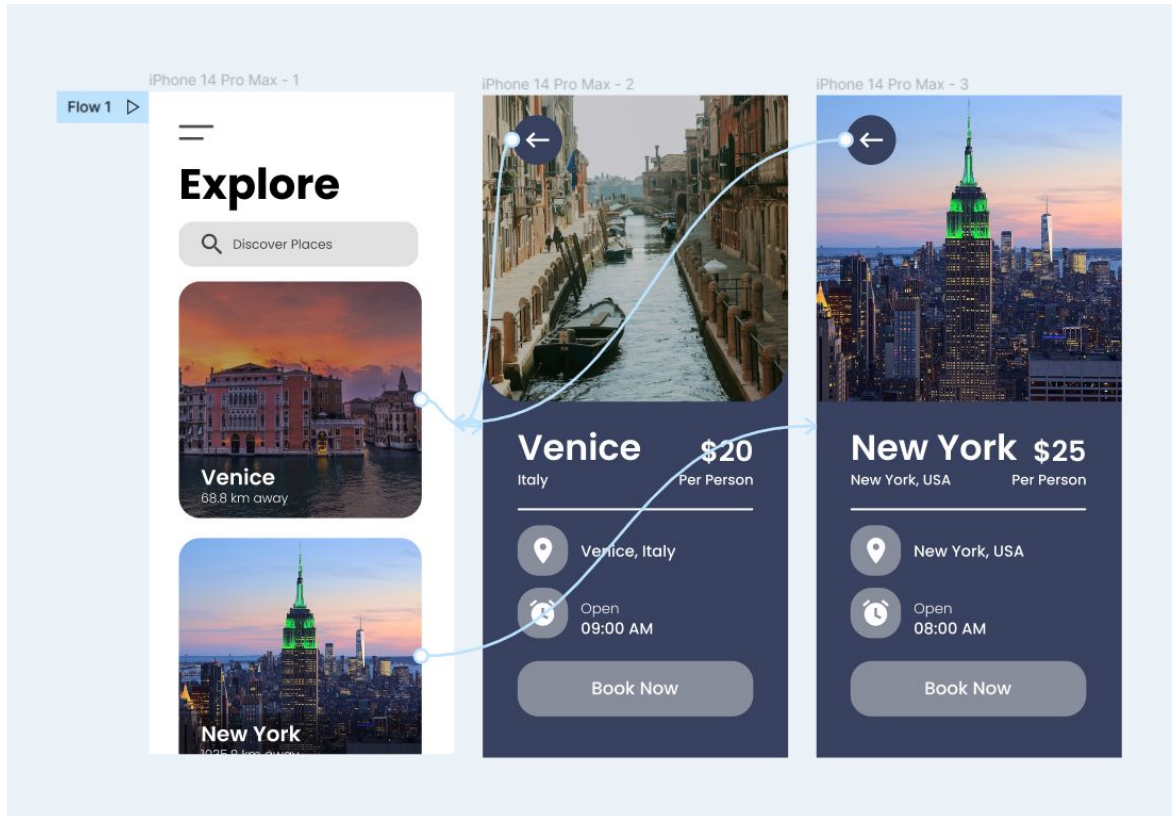


Figma

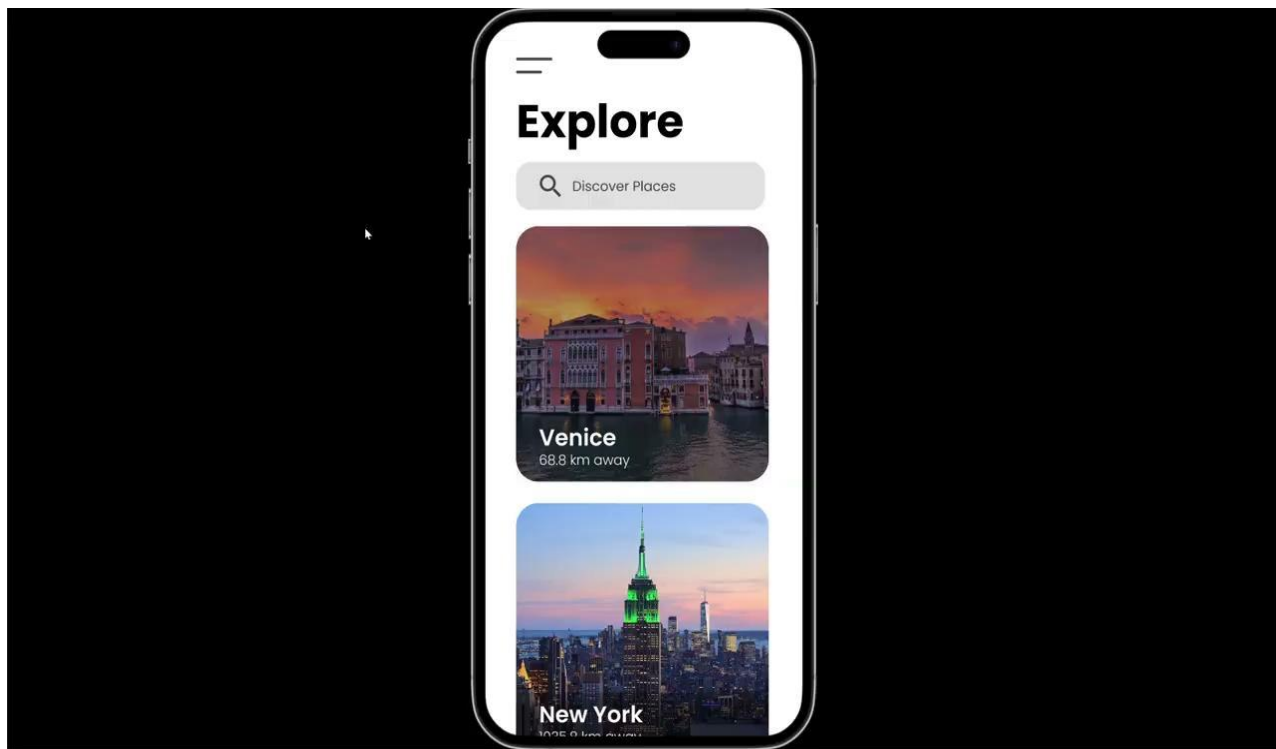


Adobe XD

Designing in progress



Design completed



Step 3: Code

Testing

Step 4: Test

```
test('given empty TravelItineraryList, user can add an item to it', () => {
  const {getByPlaceholder, getByText, getAllByText} = render(
    <TravelItineraryList />,
  );

  fireEvent.changeText(
    getByPlaceholder('Enter travel destination'),
    'New York City',
  );
  fireEvent.press(getByText('Add the item to list'));

  const nycElements = getAllByText('New York City');
  expect(nycElements).toHaveLength(1); // expect 'nyc' to be on the list
});
```

Testing (*cont.*)

Step 4: Test

```
test('An item should be added', () {  
  var destination = 'New York City';  
  locations.add(destination);  
  expect(locations.items.contains(destination), true);  
});
```

Deployment

Step 5: Deployment



Official app marketplace for iPhones, iPads, and iPod Touch devices. Offers a curated selection of high-quality, secure, and private apps. Access to millions of iOS users worldwide.



Google Play

Official app marketplace for Android devices. Vast selection of apps and games, ranging from free to paid. Access to millions of Android users worldwide.

Maintenance

Step 6: Maintain

Ongoing process of keeping app up-to-date and secure. Fix bugs, add features, address vulnerabilities. Monitor performance and user feedback. Ensures relevance and competitiveness.



Questions?

About App Dev, the Track, and Next Workshops

Reference

- React Native code can be found here:
https://github.com/appdevumd/bitcamp_reactnative
- And Flutter code can be found here:
https://github.com/appdevumd/bitcamp_flutterdemo
- Before next workshop, install Flutter and Python and make a Firebase account (instructions will be sent in the Slack)